

FCQ

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FINAL HOMEWORKS #7 & #8

HW #7 – Excel based- posted

(Portfolio Optimization & Bond Duration & Beta)

HW #8 – Calculus Review for the Final

(Pearson – now posted on Canvas)

BOTH will be due Thursday May 1st

FINAL Exam – Monday 5/5 (KOBL 100)***

Final Thoughts on Portfolio Optimization

*given weights

Stocust + RF portfolio =

$$E(R_p) = w_{\text{stocust}} E(R_{\text{stocust}}) + w_{\text{RF}} E(R_{\text{RF}})$$

$$\sigma_p = \sqrt{w_s^2 \sigma_s^2 + w_{\text{RF}}^2 \sigma_{\text{RF}}^2 + 2w_s w_{\text{RF}} \text{Covar}(s, \text{RF})}$$

Covar = 0

Bond Duration

Interest Rate / Bond Price Relationship

measure of interest
rate risk $\rightarrow$ rates $\uparrow$ prices $\downarrow$

$$\frac{dp}{dy} \quad \} \quad \Delta p \Rightarrow \text{change in price}$$



QUIZLET

Interest Rate / Bond Price Relationship

Rates ↑ price ↓

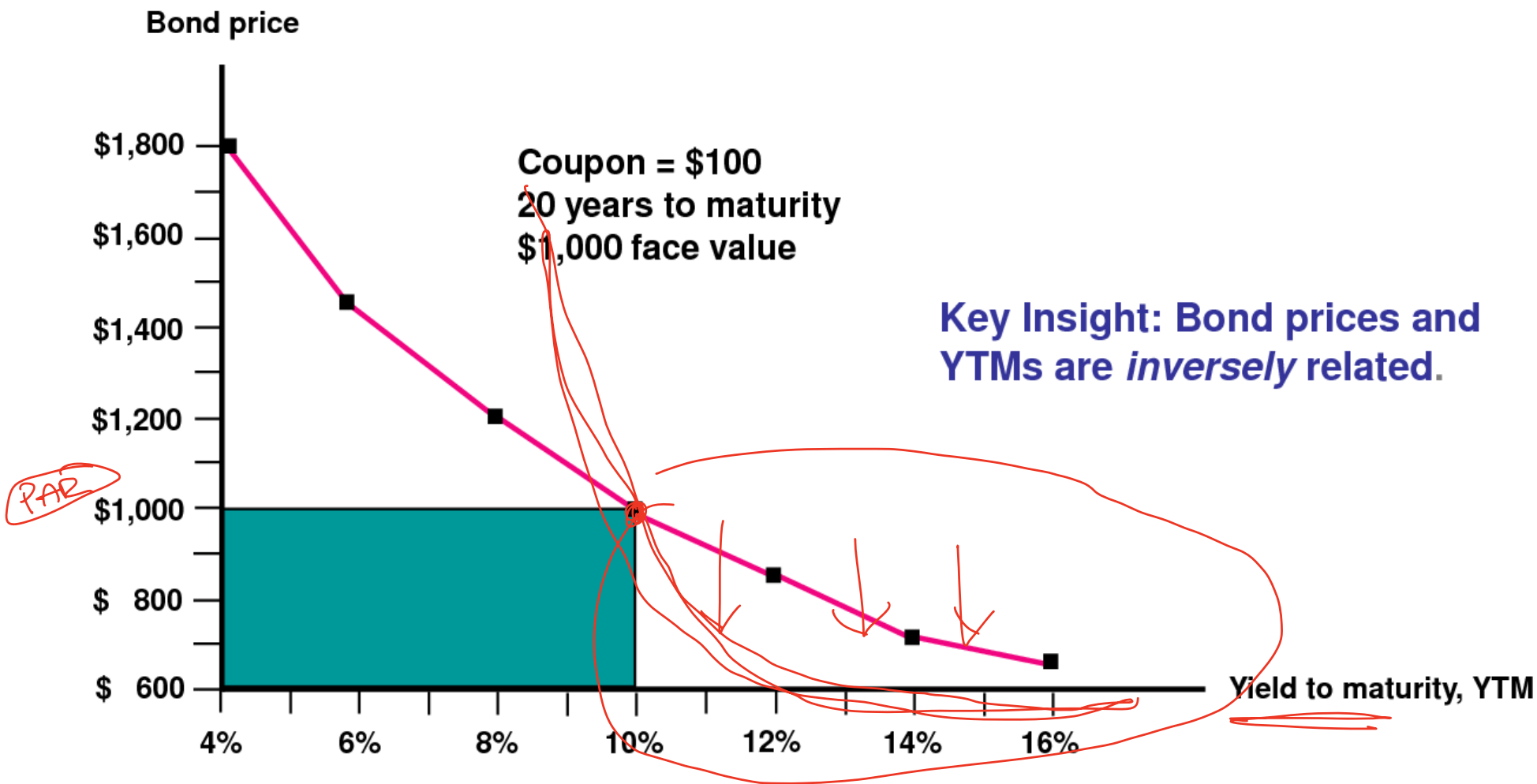
#1: When the market interest rate rises, the price of the bond falls (i.e., **it sells at a discount of par**)

#2: When the market interest rate falls, the price of the bond rises (i.e., **it sells at a premium of par**)

Rates ↓ price ↑

#3: The price of a bond will always equal par if the market interest rate equals the coupon rate

Bond Price Sensitivity to Changes in Market Interest Rates (YTM)



Interest Rate Risk on a Bond

The risk associated with a reduction in the market price of a bond resulting from a rise in market interest rates

The greater the term to maturity, the greater the change in price for a given change in market interest rates

Interest Rate Risk

What if two bonds have the same term to maturity- does that necessarily mean that they carry the same interest rate risk? (i.e., potential for a given change in price)

Assume the following two bonds:

a 20 year, 10% coupon bond

a 20 year, 6% coupon bond

⇒ *Coupon rate dictates Payment*
\$100/year
\$60/year

Which one do you think has the greatest interest rate risk (price change) for a given change in interest rates?

(LOWER coupon) → *higher duration*

Interest Rate Risk on a Bond

The risk associated with a reduction in the market price of a bond resulting from a rise in market interest rates:

The greater the term to maturity, the greater the change in price for a given change in market interest rates

A bond that will pay back more quickly to the holder (in terms of coupon cash flows) has less price risk

Bond Interest Rate Risk (EXCEL)

OPEN “BONDS duration” excel file

= PRICE(inputs



settlement maturity CPN

Yield

& redemption,



scale to investment

100 frequency basis

Concept of Bond Duration

#² math calculation

#2 excel function

Duration is the average lifetime of a debt security's stream of payments:

final!

Hw.

Duration is a more complete measure of a security's interest rate risk than maturity because **duration takes into account the timing of cash flows, as well as maturity** } semi-annual bond

the higher the duration of a bond, the higher the interest rate risk, and the more its price will drop as interest rates rise

Bond Cash Flow Review

CPN $\rightarrow$ PMT

1.50% U.S. Treasury bond with 7 years to maturity, paying interest semi-annually, with a current market price of \$1003.75, and par value of \$1000.

$$(1.5\%)(1000) = 15 \text{ year}$$

The 1.50% is the stated coupon (as a percent of par)

This bond will pay \$15.00 per year in interest

(.015 x \$1,000); **since it is a semi-annual bond, it will pay \$7.50 every six months**

At maturity, the bond will pay \$1,000 (par value) plus ~~six months interest (\$7.50)~~ or a total of \$1,007.50

today

maturity

7.50

\$1000

Duration Definition (annual)

Duration is the average lifetime of a bond's stream of payments:

$$Duration = \frac{\sum_{t=1}^n t \frac{CF_t}{(1+R)^t}}{\sum_{t=1}^n \frac{CF_t}{(1+R)^t}}$$

Handwritten annotations:

- Sum of $T \times \text{this component or (PV)}$ (pointing to the numerator)
- Sum of all the PVs or PRICE (pointing to the denominator)
- $PV = \frac{FV}{(1+r)^n}$ (written to the right)
- PRICE (pointing to the denominator)

Duration Definition (semi-annual)

Duration is the average lifetime of a bond's stream of payments:

$$Duration = \frac{\sum_{t=1/2}^n t \frac{CF_t}{(1 + R/2)^{2t}}}{\sum_{t=1/2}^n \frac{CF_t}{(1 + R/2)^{2t}}}$$

Duration Example (annual)

Compute the duration of a 6 year, 8% coupon bond if YTM is 8%. Assume annual interest payments and par value of \$1000.

Maturity	Cash flow	PV	$t \times PV$
1	80		
2	80		
3	80		
4	80		
5	80		
6	1080		

Example #2

Compute the duration of a 3 year, 6% coupon bond if YTM is 7%. Assume annual interest payments and par value of \$1000.

Maturity	Cash flow	PV	$t \times PV$
1	60		
2	60		
3	1060		

A. 2.23 years

B. 2.63 years

C. 2.43 years

D. 2.83 years

Duration Example (semi-annual)

Compute the duration of a 2 year, 10% treasury note if YTM is 5%. Assume par value of \$1000.

t	Cash Flow	PV	$t \times PV$
0.5	50		
1.0	50		
1.5	50		
2.0	1050		

$$\frac{\sum_{t=1/2}^n t \frac{CF_t}{(1 + R/2)^{2t}}}{\sum_{t=1/2}^n \frac{CF_t}{(1 + R/2)^{2t}}}$$

A. 1.62 years

B. 1.87 years

C. 1.75 years

D. 1.98 years

Using Excel to Calculate Duration

= duration()

Example for 10 year, 10% coupon bond with market rate of 10%:

Settlement: DATE(2009,2,1) Assume, Feb 1, 2009

Maturity: DATE(2019,2,1) Note: 10 years to maturity

coupon: 10% (this is the coupon yield)

Yld: 10% (this is the yield to maturity)

Frequency: 2 (assume interest is paid semi-annually)

Basis: 1 (assume basis 1 for treasuries)

Formula result = 6.54

Duration Summary

Duration is the average lifetime of a security's stream of payments:

The lower the coupon rate (maturity equal), the longer the duration

The longer the term to maturity (coupon equal), the longer duration

Zero-coupon bonds, which have only one cash flow, have durations equal to their maturity

Calculated Duration Data

Duration for a 10 year bond assuming different coupons yields:

Coupon 10%

Duration 6.54 yrs

Coupon 5%

Duration 7.99 yrs

Zero Coupon

Duration 10 yrs

Duration for a 10% coupon bond assuming different maturities:

5 years

Duration 4.05yrs

10 years

Duration 6.54 yrs

20 years

Duration 9.00 yrs

Examples

Consider two bonds – each with a maturity of 10 years. One has a coupon of 5% and the other 10%. Which has a greater duration?

The 5% coupon bond

Consider a zero coupon bond with a maturity of 5 years. The annualized yield on the bond goes from 5% to 6%. What happens to its duration?

Nothing

Risk and Duration

The following represents the relationship between duration and risk:

$$\Delta P = - \left[\frac{\Delta Y}{\left(1 + \frac{YTM}{m}\right)} \right] * (Duration) * (Price)$$

= mod duration * price

CONCEPTUALIZE THIS

where m = periods per year

Example

A 2 year, 10% semi-annual bond is priced at \$1094 and has duration of 1.87 and a YTM of 5%.

If rates increase to 6%, by how much would the price of the bond change?

A. -\$19.95

C. -\$22.45

B. -\$26.29

D. -\$27.01

$$\Delta P = - \left[\frac{\Delta Y}{\left(1 + \frac{YTM}{m}\right)} \right] * (Duration) * (Price)$$

where m = periods per year

General Duration Rule

For every 1% increase or decrease in interest rates, a bond's price will change approximately 1% in the opposite direction ***for every year*** of duration...

If a bond has a duration of 5 years and interest rates increase by 1%, the bond's price will decline by approximately 5%.

In our previous example, the bond had duration of 1.87, and price declined by 1.83%

Risk and Duration

The following represent the relationship between duration and risk:

$$\Delta P = - \left[\frac{\Delta Y}{(1 + YTM/m)} \right] * (Duration) * (Price)$$

If we define **modified duration** as $D_{mod} = \frac{(Duration)}{(1 + YTM/m)}$

We have $D_{mod} = \frac{dp/dy}{Price}$

Modified Duration

$$D_{mod} = \frac{dp/dy}{Price} \qquad \frac{dp}{dy} = (D_{mod}) * (Price)$$

$$slope = (D_{mod}) * (Price) = \frac{rise}{run}$$

Change variables, observe effect on duration

Duration & Immunization

Protect the expected return of a security by:

matching duration with planned holding period

Changes in interest rates affect price & reinvestment, but keep the security rate of return constant

Set duration = planned holding period
(5)

match duration with (5) investment period

Example

Insurance company or pension fund needs to make a guaranteed payment at a particular time in the future

Assume the following:

target payment = \$1,469

time = 5 years

We can match duration with the planned holding period... (5 years)

Duration & Immunization

when duration is equal to planned holding period, changes in interest rates affect price & reinvestment, but keep the security rate of return constant.

Therefore, the desired payout is constant, regardless of a change in rates...

Previous Duration Example

Compute the duration of a 6 year, 8% coupon bond if YTM is 8%. Assume annual interest payments and par value of \$1000.

Maturity	Cash flow	PV	$t \times PV$
1	80	74.07	74.07
2	80	68.59	137.18
3	80	63.51	190.53
4	80	58.80	235.20
5	80	54.45	272.25
6	1080	680.58	4083.48

$$\text{duration} = 4992/1000 \\ = 4.99 \text{ years}$$

Cash flow after 5 years

target payment = \$1,469

Consider the 6 year, 8% coupon bond, and YTM is 8%. Assume annual payments and par value of \$1000. If rates stay at 8%, what is cash flow after 5 years?

Maturity	Cash flow
1	80
2	80
3	80
4	80
5	80
6	1080

**coupons & reinvestment proceeds =
sale price at year five =**

TOTAL =

What if rates go down?

target payment = \$1,469

Consider the 6 year, 8% coupon bond, and YTM is 8%.

If rates drop to 7%, what is cash flow after 5 years?

Maturity	Cash flow
1	80
2	80
3	80
4	80
5	80
6	1080

coupons = $(\$80 \times 5) =$

sale price at year five =

reinvestment proceeds =

TOTAL =

What if rates go up?

target payment = \$1,469

Consider the 6 year, 8% coupon bond, and YTM is 8%.

If rates rise to 9%, what is cash flow after 5 years?

Maturity	Cash flow
1	80
2	80
3	80
4	80
5	80
6	1080

coupons = $(\$80 \times 5) =$

sale price at year five =

reinvestment proceeds =

TOTAL =

